Anna Ling

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EDUCATION

Rice University | *Houston, TX* Expected May 2029

B.S. Mechanical Engineering, Computational Engineering Specialization

- Minors: Computational and Applied Mathematics, Spanish | GPA: 3.47/4.00 | National Merit Scholar | SAT: 1540
- Relevant Coursework: Engineering Computation (Python & MATLAB), Differential Equations & Linear Algebra, Multivariable Calculus, Engineering Design Tools
- Activities: American Society of Mechanical Engineers, Society of Women Engineers, Duncan College

TECHNICAL SKILLS

Languages: Python, SQL, MATLAB, HTML/CSS
Libraries & Frameworks: NumPy, Matplotlib, Flask
Tools & Data: Git, SQLite, Excel, Google Sheets, Airtable, PitchBook, SolidWorks (FEA)
Concepts: Signal processing (FFT, spectrograms), data cleaning & validation, relational databases, REST APIs, Bayesian averaging

TECHNICAL EXPERIENCE

Rice University Wireless Lab | *Undergraduate Researcher* Aug 2026 – Present

- Building a Python pipeline that converts motion-capture body meshes (AMASS, SMPL) into simulated 77 GHz mmWave radar signals
- Generating labeled synthetic radar data with range-Doppler FFT and micro-Doppler processing to train machine-learning human motion models
- Collaborate with a PhD mentor under Prof. Ashutosh Sabharwal (ECE), translating wireless sensing research questions into simulation code

Rice University School of Engineering and Computing | *Data Validation Assistant* Oct 2026 – Present

- Clean and validate company and founder data for 924+ startups in a university startup dataset, cross-referencing PitchBook and public records
- Detect duplicate, inconsistent, and unverifiable entries in Excel, Google Sheets, and Airtable, documenting data provenance for every validated field
- Compile structured venture capital data, including funding stage, round sizes, and valuations, to support research on university-affiliated startups

Rice University Formula SAE | *Chassis Subteam Member* Jan 2026 – Present

- Model chassis frame components in SolidWorks, running FEA and hand stress calculations to verify FSAE structural equivalency requirements
- Compared carbon fiber and steel using Young's modulus and stress analysis, making independent material decisions on a 12-member subteam

U.S. Merchant Marine Academy | *Research Assistant, Biochemistry Dept.* Oct 2023 – Aug 2024

- Performed data analysis and authored manuscript sections for silver nanoparticle water purification research under Prof. Ping Furlan

TECHNICAL PROJECTS

Sushi Finder | *Python, Flask, SQL (SQLite), Google Places API, HTML/CSS, Git* Sep 2026

- Built a Flask web app integrating the Google Places REST API, ranking restaurants with a Bayesian-adjusted rating to correct low-review inflation
- Added a SQLite database to store retrieved restaurant data and wrote SQL queries to analyze the stored results

COMMUNICATION & LEADERSHIP EXPERIENCE

Brown University Pre-College Programs | *STEM Hybrid Assistant* Jun 2026 – Jul 2026

- Collaborated with a STEM program team across 3 summer sessions, supporting high school students in computational physics, biology, computer science, and algebra

Pub at Rice | *Bartender* Aug 2026 – Present

- Communicate with high volumes of customers each shift while managing sales and maintaining accurate financial records at a student-run pub

Kumon North America | *Tutor* Jan 2025 – Aug 2025

- Explained math and reading concepts one-on-one to students ages 7–14, tailoring instruction and tracking each student's weekly progress

Spoken Languages: English (native), Spanish (proficient)